

Gate E1 report — run_v1.22_r1 (Paper IV v1.22-r1)

Verdict: **PASS**. Evidence: 20_EVIDENCE/E1/. Auditor: Claude Opus 5.5, same session as run_v1.2_r1/run_v1.21_r1; no Lean executed.

E1 — Statements, constants and tags (regression)

The basis is the auditor's own diff (20_EVIDENCE/E2/auditor_diff_*.diff) and E6_PARITY.json.

- No theorem, proposition, lemma or corollary statement changed in either language; the theorem-heading line counts are unchanged (33 EN, 34 ES).
- Equation tags are identical to v1.21 in both languages, and no display formula was removed. The 7 added displays are the C.3 derivation (EN and ES identical).
- The final constants are unchanged: $k_0 = 3 \cdot 10^{19} + 1$, h , $N = T(h + 7)$, $b = N^2 \leq T(h + 8)$, c_{poly} , c_{far} , c_T , $P(s)$, γ_s , A_s and the thresholds of (F.1). C.3 now derives $T(h + 7)$ (E2), and the value is not modified.
- NEW-01 (EN 5.1 heading, A.2 row, ES Lemma 3.2 heading) is fixed. The remaining ES §7.2 Model formatting residual is recorded under E6.
- X-17 and X-18 are kept as OBSERVATIONS (unchanged; the author made no change).

E1 verdict: PASS.

PDF generated by pandoc (HTML) + PyMuPDF Story from the Markdown source; not compiled from TeX. The TeX file alongside is pandoc's standalone conversion of the same Markdown.